

Identified Problem

- RPW infestation – one of the major threats to coconut cultivation in Sri Lanka
- Problems with the device that is currently in use
 - Hard to distinguish RPW sounds manually
 - Captures too much noise
 - Usage of earphones is inconvenient

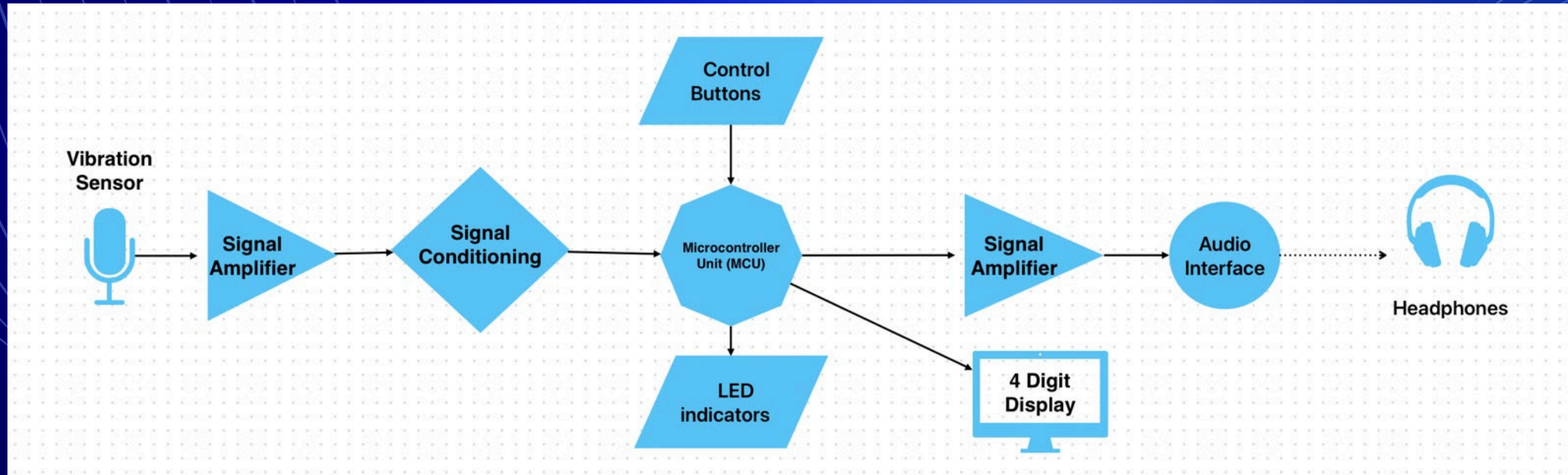


Goal of our product

- To capture an audio signal with less noise
- To make the detection process convenient and easy for the user



Design Architecture



- Acoustic Sensor with high sensitivity
- Non- inverting amplifier with control gain
- Machine Learning model running locally on ESP32

Piezo sensor

Attached to the trunk
Internal vibrations of the tree trunk are detected and transferred to the circuit

Volume Controller

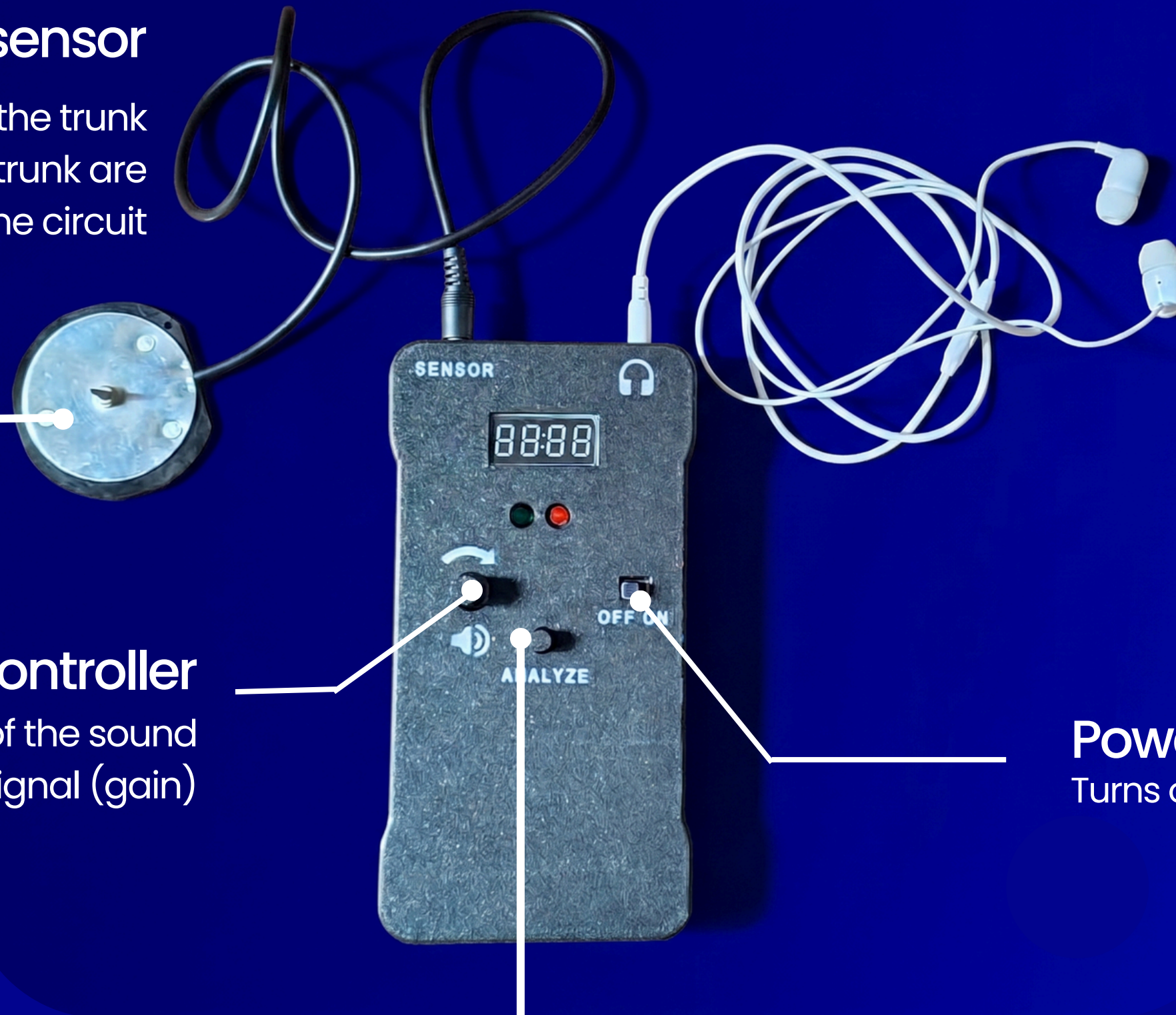
Controls the volume of the sound signal (gain)

Push button

To start the model running

Power Switch

Turns on the device





TEAM IOTRIX

7 segment display

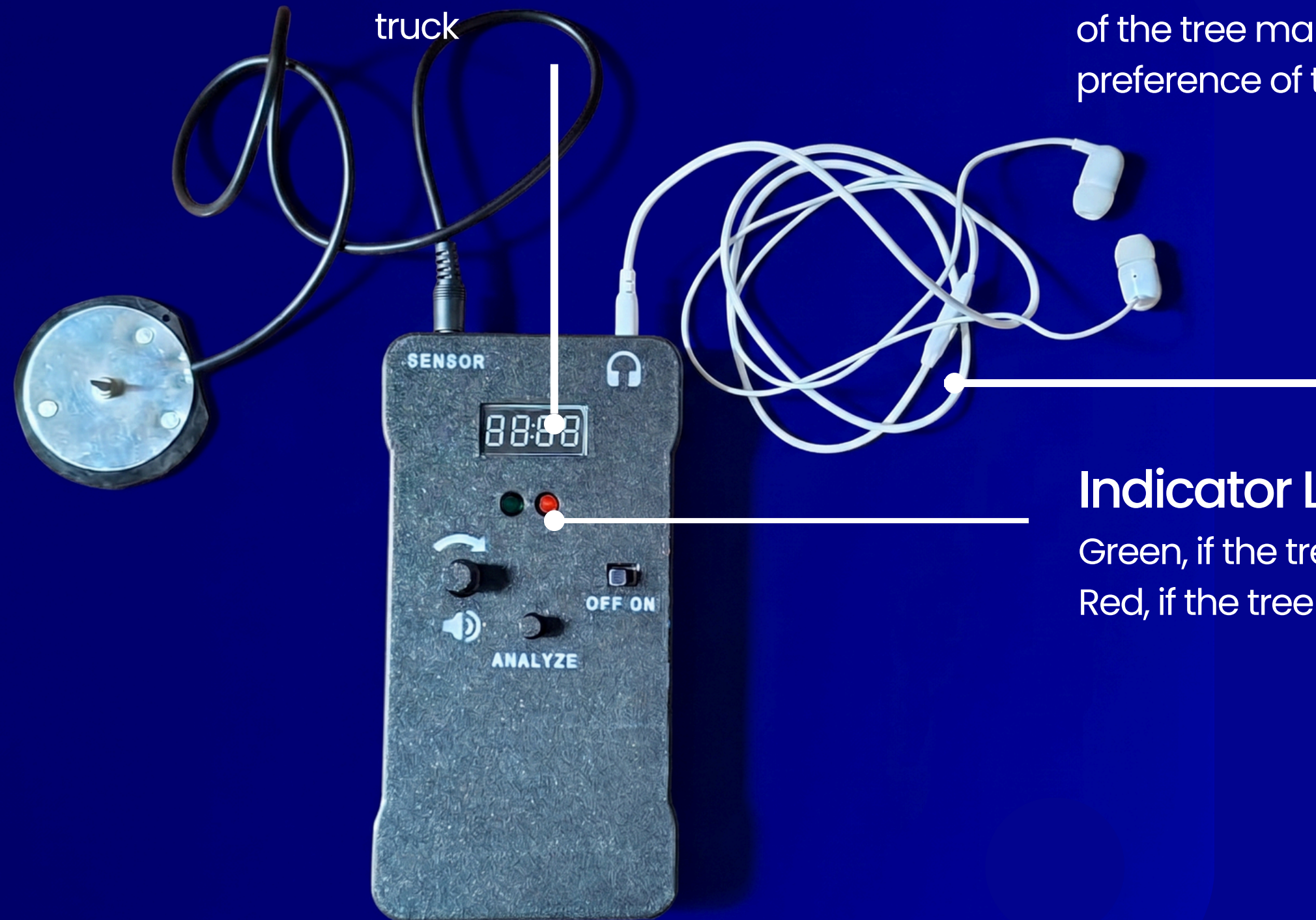
Displays the probability of the presence of a RPW inside the truck

Earphones

To listen to the internal vibrations of the tree manually, on the preference of the user

Indicator LEDs

Green, if the tree is healthy
Red, if the tree is infected



Task Allocation Among Members

1) Pasindu

- Designed the hardware setup of the circuit.
- Contributed to the design and training of the ML model.
- Designed the schematics and the PCB layout.
- Contributed to circuit testing and implementation.
- Participated in ideation, user research, and analysis.

2) Pasan

- Designed the PCB layout.
- Carried out soldering and assembly.
- Contributed to the design and training of the ML model.
- Contributed to circuit testing.
- Participated in ideation, user research, and analysis.

3) Dulshi

- Contributed to train the ML model.
- Designed the enclosure.
- Contacted relevant resource persons to carry out testing.
- Contributed to circuit testing and implementation.
- Participated in ideation, user research, and analysis.

4) Dulani

- Designed the enclosure.
- Contacted relevant resource persons to carry out testing.
- Contributed to circuit testing and implementation.
- Participated in ideation, user research, and analysis.